AI-Integrated Metacognition (AIMC) Framework

Reconceptualizing Metacognition in AI-Augmented Learning Environments

Level 3: Independent Metacognition

(Without-AI Transfer)

L3

- Self-regulated learning without AI support
- Internalized monitoring & evaluation skills
- Transfer to novel contexts (Cognitive Dependency Test)

↓ *Internalization*

Level 2: Meta-AI Awareness

(About-AI Knowledge)

L2

- Understanding AI capabilities & limitations
- Evaluating AI output reliability & accuracy
- Knowing when to use/ not use AI assistance

↓ *Reflection*

Level 1: AI-Assisted Metacognition

(With-AI Context)

L1

- Prompt engineering as planning
- Output evaluation as monitoring
- Iterative refinement as self-regulation

Key Finding

Current evidence primarily assesses Level 1. The Cognitive Dependency Hypothesis predicts divergent effects at Level 3 (independent metacognition transferable to AI-absent contexts).

You, H. (2026). Generative AI in Higher Education: A Three-Level Meta-Analysis. Pennsylvania State University